

Launch instances *h-slv1* and *h-slv2* serially using Hadoop-Base as the instance boot source. Launching of the first instance from a snapshot is pretty slow. In case the IP addresses are not the same as your guess in */etc/hosts*, edit */etc/hosts* on each of the three nodes to the correct value. For your convenience, you may want to make entries for *h-slv1* and *h-slv2* on the desktop */etc/hosts* file as well.

The following commands should be run from Fedora on *h-mstr*. Reformat the *namenode* to make sure that the single node tests are not causing any unexpected issues:

```
$ sudo runuser hdfs -s /bin/bash /bin/bash -c "hadoop
namenode -format"
Start the hadoop services on h-mstr.
$ sudo systemctl start hadoop-namenode hadoop-datanode
hadoop-nodemanager hadoop-resourcemanager
```

Start the *datanode* and *yarn* services on the slave nodes:

```
$ ssh -t fedora@h-slv1 sudo systemctl start hadoop-datanode
hadoop-nodemanager
$ ssh -t fedora@h-slv2 sudo systemctl start hadoop-datanode
hadoop-nodemanager
```

Create the *hdfs* directories and a directory for user *fedora* as on a single node:

```
$ sudo hdfs-create-dirs
```

```
$ sudo runuser hdfs -s /bin/bash /bin/bash -c "hadoop fs
-mkdir /user/fedora"
$ sudo runuser hdfs -s /bin/bash /bin/bash -c "hadoop fs
-chown fedora /user/fedora"
```

You can run the same tests again. Although you are using three nodes, the improvement in the performance compared to the single node is not expected to be noticeable as the nodes are running on a single desktop.

The *pi* example took about one minute on the three nodes, compared to the 90 seconds taken earlier. Terasort took 7 minutes instead of 8.



Note: I used an AMD Phenom II X4 965 with 16GB RAM to arrive at the timings. All virtual machines and their data were on a single physical disk.

Both OpenStack and Mapreduce are a collection of interrelated services working together. Diagnosing problems, especially in the beginning, is tough as each service has its own log files. It takes a while to get used to realising where to look. However, once these are working, it is incredible how easy they make distributed processing! 

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